A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 32

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 2 points below. **1** Single genes can be transferred from any species, not just closely related ones that produce fertile hybrids **1** Desired characteristics introduced without linked unwanted ones – faster **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'GM crops are unnatural' as an environmental impact - credit gene flow, resistance, effects on non-target species or reduced pesticide use	2
01.2	**1 mark each** 1 mark for each of the 3 points below. **1 Bt:** reduces insecticide spraying so fewer non-target deaths, lower cost / energy **1 Roundup Ready:** weeds controlled by spraying glyphosate without harming the crop, so higher yields with less cultivation / tillage **1 Golden rice:** provides vitamin A in the staple diet, reducing about 500 000 cases of blindness a year in India / sub-Saharan Africa **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
01.3	**1 mark each** 1 mark for each of the 4 points below. **1** Gene transfer by pollen to conventional / organic crops or wild relatives (eg Bt gene killing insects feeding on wild plants) **1** Antibiotic-resistance marker genes could transfer to pathogenic bacteria – possible allergies – gene transfer to gut bacteria **1** Detection in the human food chain from crops approved only for animal feed **1 Economic:** seeds must be bought each year from patent-holding companies, raising costs and reducing local crop diversity and seed saving in LEDCs – herbicide-resistant crops may increase herbicide use **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts • 'GM crops are unnatural' as an environmental impact - credit gene flow, resistance, effects on non-target species or reduced pesticide use	4
01.4	1 mark for the point below. **1** Public / political concern that risks are not fully assessed - more evidence needed (precautionary principle) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'GM crops are unnatural' as an environmental impact - credit gene flow, resistance, effects on non-target species or reduced pesticide use • the precautionary principle described as banning anything new - it shifts the burden of proof onto showing that harm is unlikely	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 323-324; AQA spec 3.5.1.2.	
02.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** Environmental DNA (eDNA) analysis	1
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Transmitters are too heavy for small birds / their batteries would not last for a long-term study **1** Geolocator tags are small and only record time and light levels – latitude and longitude for each day are worked out when the bird is recaptured and the data downloaded **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Each separate (regional) population has its own unique gene pool **1** DNA from seized ivory can be compared with the database to identify where the elephant came from, so poaching hotspots / trade routes can be targeted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 404-405; AQA spec 3.1.2.3.6.	
03.1	**1 mark each** 1 mark for each of the 2 points below. **1** Small seeds, genetically uniform, high and rapid germination rate so results are quick and reliable **1** Wild species have different ranges of tolerance so cress results cannot be applied directly **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliable** → repeatable / gives consistent results when repeated • **tolerance** → the range of conditions in which a species can survive **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Preliminary study with widely spaced pH values to find where germination occurs **1** Final range from below the lowest pH giving germination to above the highest, with closer intervals **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.3	**1 mark each** 1 mark for each of the 4 points below. **1** Temperature (constant, natural germination temperature) **1** Light, water volume, number of seeds per dish, seed source **1** Growth medium (filter paper; soil would alter pH) – nutrients not needed for germination **1** Continue until it is clear that ungerminated seeds will not germinate, so that a delay in germination is not mistaken for failure **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4

Do not write outside the box

Question	Marking guidance	Marks
03.4	**1 mark each** 1 mark for each of the 2 points below. **1** Surface water affects fresh mass and drying damages the plants – dry mass requires killing the plants so the same plants cannot be re-measured **1** Leaf area is a reasonable estimate of biomass that allows repeated measurement of the same plants over time **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 416-418; AQA spec 3.7.2.2.	
04.1	**Any 2 of 6** 1 mark each for any 2 of the 6 points below, in any order, maximum 2. **1** Potato bean (N America): high protein content **1** Morama bean (southern Africa): protein content similar to soya, drought resistant **1** Yeheb tree (Somalia): edible nuts, drought resistant, grows on poor soils **1** Perennial maize (Mexico) / Kernza: no need to buy seed each year, less ploughing so less soil erosion **1 American bison:** graze grassland too poor for cattle **1 Common eland:** browse and graze so can be kept with cattle without severe competition **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	2
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Indigenous species are better adapted to local conditions such as climate, pests / diseases and soil **1** So they have higher survival rates / productivity is not reduced, as it is when introduced species are kept in areas to which they are not adapted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.3	1 mark for the point below. **1** They grow in soils with a high salt content, so their genes could be used in breeding programmes to increase the salt tolerance of crops grown on soils salinised by irrigation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **tolerance** → the range of conditions in which a species can survive	1

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 11; AQA spec 3.1.2.1.1.	
05.1	**1 mark each** 1 mark for each of the 3 points below. **1** Tropical intertidal coastal forests of halophytic (salt-tolerant) trees in saline, oxygen-deficient (anaerobic) mud **1** Prop / aerial roots for support in soft sediment and for gas exchange (pneumatophores) – salt excretion **1** Seeds germinate on the parent tree and float, rooting where they touch sand - rapid natural regeneration **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
05.2	**1 mark each** 1 mark for each of the 4 points below. **1 Coastal erosion / storm protection:** roots and vegetation absorb wave energy (least tsunami damage in 2004 where mangroves were intact) **1 Fisheries:** nursery grounds where dense roots protect young fish that later live on reefs / open sea **1 Protection of coral reefs:** slow river water so suspended solids settle and nutrients are absorbed before reaching reefs **1 Resources:** timber, fuel – medicines (antimicrobials against MRSA, cyanobacterium for lung cancer) – high biodiversity (crabs, shrimps, sponges, birds) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
05.3	**1 mark each** 1 mark for each of the 3 points below. **1** Clearance for aquaculture (shrimp ponds), urban development, ports, tourism – pollution – sea level rise / climate change – destruction of coral reefs that shelter the coast **1** Legal protection / designated areas (eg Sri Lanka protected all remaining mangroves in 2015) **1** Replanting and natural regeneration – control of damaging activities – Ramsar sites **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 66-67; AQA spec 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**1 mark each** 1 mark for each of the 4 points below. **1** Natural processes are driven by renewable, low energy-density solar energy captured by photosynthesis – human processes usually use high energy-density non-renewable fossil fuels **1** Natural processes occur at low temperatures because enzymes lower the activation energy of reactions – human processes need high temperatures and pressures **1** Natural processes produce no toxic wastes / their products are raw materials for other processes – industrial processes emit CO2 and NOx **1** So natural fixation is sustainable indefinitely with a low carbon footprint **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	4
06.2	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Hydrological cycle (evaporation, precipitation) **1** Carbon cycle / photosynthesis **1** Atmospheric circulation / winds **1** Thermohaline circulation (density differences from heating / evaporation) **1** Decomposition **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Thermohaline** → the ocean conveyor driven by temperature and salinity **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'plants breathe in carbon dioxide and breathe out oxygen' - name photosynthesis and respiration	3
06.3	**1 mark each** 1 mark for each of the 3 points below. **1** Bioleaching and phytomining extract metals at low temperature instead of high-temperature smelting **1** Anaerobic digestion and composting treat wastes – enzymes in industry (biological washing powders, biofuels from algae / microorganisms) **1** Heat pumps and solar thermal capture low-density heat – low-temperature manufacturing using biological polymers – fuel cells instead of combustion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2.	
07.1	**1 mark each** 1 mark for each of the 4 points below. **1** Sigmoidal (S-shaped) growth curve **1** Slow start (few breeding individuals), then rapid / exponential growth (years 4-10) as abundant food and space mean birth rate far exceeds death rate **1** Growth slows as density-dependent factors (intra-species competition for food, disease) raise the death rate **1** Population stabilises / fluctuates around about 920-950 - the carrying capacity - where birth rate equals death rate **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **carrying capacity** → the largest population the area can support **Do not credit** • 'the population cannot grow any more' as a definition - numbers fluctuate around the carrying capacity	4
07.2	**1 mark each** 1 mark for each of the 2 points below. **1 Density-dependent:** effect on survival increases with population density - food shortage / intra-species competition, disease spread, predation **1 Density-independent:** chance of survival unaffected by density - drought, flood, volcanic eruption, severe winter **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Without wolves (natural predators removed) the population is regulated only by starvation / disease at a high density **1** At this density deer over-graze, preventing tree regeneration and damaging habitats / other species – culling keeps numbers below the damaging level (as in Scotland) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.4	**1 mark each** 1 mark for each of the 2 points below. **1** Rising prey numbers give predators more food so predator numbers rise – high predation then reduces the prey population **1** Food shortage reduces predators, allowing prey to recover - cyclic fluctuations with predators lagging behind prey **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 78-81; AQA spec 3.1.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**1 mark each** 1 mark for each of the 2 points below. **1** Island communities evolved from few colonising species in isolation, often without mammal predators or large herbivores **1** So indigenous species lack adaptations (eg ground-nesting birds, flightlessness) to survive introduced cats, rats, pigs, goats **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
08.2	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Competitor:** grey squirrel (North America) out-competes red squirrel - digests acorns, larger, wins nest sites – rhododendron shades ground and releases toxins preventing native regrowth **2 Predator:** American mink (escaped fur farms) reduced water vole range – cane toads in Australia – cats / rats / pigs on islands eating ground-nesting birds – Nile perch in Lake Victoria ate cichlids to extinction **2 Pathogen:** squirrel pox virus carried by grey squirrels kills red squirrels – crayfish plague carried by signal crayfish kills white-clawed crayfish – Dutch elm disease, ash dieback, sudden oak death imported on plants / soil **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	6
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** Closely related introduced species (sika deer; domestic cats) cross-breed producing fertile hybrids **1** The natural gene pool is changed / diluted with genes it would not naturally contain, so the pure indigenous species may disappear **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 24-25; AQA spec 3.1.2.2.	
09.1	**Marks shared** Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2 Rotation:** each crop has its own pests – growing a different crop each year (4-5 year cycle) means pests left at the end of a season die before that crop returns, so populations cannot build up early **up to 2** Beetle banks / hedgerows provide habitat and overwintering sites for natural predators (ground beetles, ladybirds) that eat aphids **up to 2** Rotation also evens out nutrient demands and includes legumes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4

Do not write outside the box

Question	Marking guidance	Marks
09.2	**1 mark each** 1 mark for each of the 3 points below. **1** Integrated control gives almost the same yield as chemical control (18.1 vs 18.5) with far fewer applications (1-2 vs 6) **1** It retains most of the biodiversity of cultural control (26 vs 31 species) whereas chemical control alone leaves only 8 **1** Cultural control alone loses about 12% yield – integrated control combines cultural methods, resistant varieties and carefully timed specific non-persistent pesticides only when essential - best balance of yield, cost and impact **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity • **persistent** → does not break down / stays in the environment for a long time **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	3
09.3	**1 mark each** 1 mark for each of the 3 points below. **1** Large numbers of males are sterilised with gamma radiation and released **1** Females of many insects mate only once and store sperm, so mating with a sterile male produces no offspring – enough sterile males reduce fertile matings below replacement level (screw worm fly, Mediterranean fruit fly) **1** Only the target species' females mate with them so non-target species are unaffected – fails if sterilised males behave abnormally or re-colonisation occurs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 314-318; AQA spec 3.5.1.1.	
10.1	1 mark for the point below. **1** Development that meets the needs of the present generation without compromising the ability of future generations to meet their own needs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	1

Do not write outside the box

Question	Marking guidance	Marks
10.2	**1 mark each** 1 mark for each of the 4 points below. **1** Negative feedback resists change / returns the system towards its previous state **1** eg increased temperature increases cloud cover and albedo, or higher CO_2 increases photosynthesis **1** Positive feedback increases the change / makes it self-sustaining **1** eg melting permafrost releasing methane, declining albedo as ice melts, forest and peat fires **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **negative feedback** → a change that cancels itself out / a self-correcting cycle • **positive feedback** → a change that makes itself larger / a self-reinforcing cycle • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back **Do not credit** • 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	4
10.3	1 mark for the point below. **1** Many species with a complex food web so the loss of one has a small effect – agroecosystems have few species and lack detritivores, pollinators and pest predators that maintain stability **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1

Do not write outside the box

Question	Marking guidance	Marks
10.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Principles of natural systems:** dynamic equilibria and negative feedback – diverse systems resistant to change – low energy-density renewable energy (solar-driven hydrological, carbon and nitrogen cycles) – low-temperature processes using enzymes – material cycles with wastes as raw materials – small number of abundant elements – biodegradable non-toxic wastes • **Human systems:** non-renewable energy, high temperatures (Haber process, incineration, manufacturing), linear resource use, dispersal of minerals and mixtures that are hard to separate, persistent toxins (PCBs, organochlorines, toxic metals in electronics) • **Applications:** renewable energy and heat pumps, bioleaching, recycling and design for re-use, industrial symbiosis, organic and integrated agriculture, retaining hedgerows and pollinators, circular economy • Human responses that override feedback (fertilisers, irrigation, pesticides) and positive feedback triggered by human activity (permafrost, methane hydrates, fires) – tipping points • **Barriers:** population size and affluence, attitudes of urban populations to distant resources, economics, need for international cooperation, existing infrastructure • Justified conclusion on how far natural principles can be applied **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9
	• **sustainable** → meeting present needs without stopping future generations meeting theirs • **negative feedback** → a change that cancels itself out / a self-correcting cycle • **biodegradable** → broken down by decomposers **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	
	Model answer - Level 3 (8 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Natural systems have remained sustainable because they run on low energy-density renewable energy, mainly sunlight driving the hydrological, carbon and nitrogen cycles; their reactions occur at low temperatures because enzymes lower activation energies; materials cycle so the waste of one process is the raw material of the next; only a few abundant elements are used; and wastes are biodegradable and non-toxic. Negative feedback keeps them in dynamic equilibrium and high diversity makes them resistant to change.	
	Human systems contrast on every point: non-renewable fossil fuels, high-temperature processes such as the Haber process and incineration, linear use that disperses minerals and mixes materials that cannot be separated, and persistent toxins like PCBs and the metals in electronics. We also override negative feedback - when degraded soil reduces yields we add fertiliser, irrigation and pesticides - and trigger positive feedback such as permafrost melting.	
	Copying natural principles is already possible in many areas: renewable energy and heat pumps, bioleaching at low temperature, recycling and design for disassembly, industrial symbiosis, legumes and organic matter instead of manufactured nitrate, and retaining hedgerows for pest predators - the basis of the circular economy. The limits are population size and affluence, the attitude of urban consumers who never see depleted resources, cost, and the need for international cooperation. So the principles can make individual activities far more sustainable, but only a large-scale, cooperative shift in lifestyle would make the whole human system sustainable.	
	Weaker answer - Level 2 (5 marks)	
	Natural systems use renewable energy from the Sun and recycle materials, so nothing runs out and there is no waste. Humans use fossil fuels which will run out and produce waste that is toxic. Natural processes happen at low temperatures but the Haber process needs high temperatures. Humans could copy natural systems by using renewable energy like solar and wind, recycling more and using less fertiliser. This would make human activities more sustainable. However it would be expensive and people would have to change their lifestyle.	
	What makes the difference	
	• Level 2 states the contrast correctly but does not explain the mechanisms (enzymes, negative feedback, diversity, dispersal of minerals) that Level 3 uses.	
	• Level 3 applies the principles to specific technologies (bioleaching, heat pumps, industrial symbiosis, legumes); Level 2 stays at 'recycle more'.	
	• 'Evaluate the extent' - Level 3 distinguishes individual activities from the whole system in its conclusion; Level 2's evaluation is a single sentence about cost.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-374; AQA spec 3.6.1-3.6.3.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Fishing is the last large-scale human hunting activity. Discuss the environmental impacts of fishing and evaluate the methods used to make the exploitation of fish populations sustainable.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Marine productivity:** light, nutrients, upwelling – where fisheries occur • **Methods:** demersal / pelagic trawling, purse seining, drift nets, long lines, shellfish traps - effectiveness, selectivity, energy, impacts • **Impacts:** overfishing (r vs k selection, catchable vs breeding size), changed age structure, by-catch (albatross, porpoise, turtles, shrimp), ghost fishing, seabed / reef / seagrass damage, dynamite, food web effects (sandeels, tritons) • **Population dynamics and MSY:** Russell formula, difficulties of data, indicators from catches (CPUE, mean size, age), research on eggs and larvae • **Management:** quotas, mesh design, escape panels, TEDs, pingers, hook design, effort limits, method bans, NTZs, closed seasons, size limits, protected individuals, population seeding, biodegradable gear – EU CFP, IWC, CCAMLR • Evaluation with examples (cod, orange roughy, herring recovery, NTZ spillover) – role of aquaculture	25
	– judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-351; AQA spec 3.5.2.	

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Question	Marking guidance	Marks
11.2	**Discuss the extent to which the resources and services provided by other species justify the conservation of biodiversity.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Resources:** wood, fibres, oils, fuels, food – new food species (morama bean, yeheb, perennial maize, Kernza, bison, eland) and their adaptations • **Biomimetics:** wing-tip feathers, whale tubercles, shark skin, termite mounds, bird bones, gecko adhesion, burrs, lotus, spider silk • **Medicines:** poppies, cinchona, willow, yew (taxol), Mexican yam, sponges (AZT), sharks – few species yet studied – physiological research (marsupials, hippo, sponges) • **Genetic resources:** crop wild relatives, Vavilov centres, gene pools and inbreeding • **Ecosystem services:** atmospheric regulation, hydrological cycle, biogeochemical cycles, soil maintenance – interspecies relationships - food, pollination (Darwin's orchid), seed dispersal, habitat provision • **Evaluation:** economic vs intrinsic / moral arguments – services unnoticed because of dynamic equilibrium – unknown future value – justified conclusion	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	

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